

Kumar Abhishek

☎ +91 96080 13812 ✉ abhishekkumar.ch2607@gmail.com 🔗 LinkedIn 🌐 GitHub 📁 Portfolio

PROFESSIONAL SUMMARY

Electronics and Communication Engineering student with hands-on experience in **Machine Learning**, **Deep Learning**, and **Generative AI**. Experienced in building production-ready AI systems using **PyTorch**, **FastAPI**, and modern AI frameworks, with projects spanning deep learning, signal processing, LLM applications, and intelligent automation. Passionate about developing AI solutions for real-world problems across computer vision, deep learning, and generative AI.

EDUCATION

• Indian Institute of Information Technology, Una <i>B.Tech, Electronics & Communication Engineering</i>	<i>Jul 2023 – Jul 2027</i> CGPA: 7.74/10
• Delhi Model Public School, Delhi <i>Class XII (CBSE)</i>	<i>Jun 2022</i> Score: 85%

PROJECT EXPERIENCE

• InterviewBot – AI-Powered Interview Preparation Platform <i>FastAPI LangChain LangGraph Google Gemini ChromaDB MongoDB </i>	<i>Aug 2024 – Oct 2024</i> Github Live
- Built a production-ready AI interview platform comprising multiple FastAPI microservices with LangGraph workflows supporting RAG chat, interview setup, contextual assistance, and personalized interview evaluation. - Implemented RAG pipeline with ChromaDB vector store and all-MiniLM-L6-v2 Sentence Transformers for semantic search over website-indexed content; built a resume parser (PDF/DOCX via LangChain) that extracts skills and generates personalized interview questions. - Delivered MCQ test engine and mock interview module with JWT-secured FastAPI backend, async MongoDB (Motor) storage, voice input via React Speech Recognition, and auto-generated PDF performance reports using ReportLab.	
• Deep Learning for ECG Signal Analysis and R-Peak Detection <i>PyTorch Deep Learning NumPy SciPy Scikit-learn PyWavelets</i>	<i>Sep 2025 – Dec 2025</i> Github
- Implemented and benchmarked 6 R-peak detection approaches on the MIT-BIH Arrhythmia Database – spanning classical signal processing (Pan-Tompkins, Wavelet Transform) and deep learning models (1D CNN, LSTM, IncRes-UNet/RPNet, Distance Transform Learning). - Achieved 99.5% sensitivity with the Pan-Tompkins algorithm and 99.7% sensitivity / 99.6% F1-score with the Inception-Residual UNet (RPNet), employing distance transform regression for improved gradient flow and robustness to noisy signals. - Designed and trained an Inception-Residual U-Net (RPNet) in PyTorch using a distance transform regression objective, enabling robust feature learning and accurate R-peak localization in noisy ECG signals.	
• MediCure – AI-Powered Healthcare Assistant <i>FastAPI PyTorch Transformers Google Gemini React Tailwind CSS</i>	<i>Jan 2025 – Jun 2025</i> Github Live
- Fine-tuned BERT Base Transformer models in PyTorch for domain-specific medicine understanding across usage prediction, side-effect analysis, and substitute recommendation. - Developed three BERT-based classification systems covering 583 usage classes, 1,271 side-effect classes, and 43,297 medicine substitute classes, using task-specific classification heads and label encoding. - Integrated the fine-tuned Transformer models into a FastAPI inference backend for real-time medicine analysis, alongside a Google Gemini conversational healthcare assistant. - Built an end-to-end React and Tailwind CSS application exposing Transformer inference and generative AI capabilities through REST APIs.	

TECHNICAL SKILLS

- **Programming:** Python, SQL.
- **AI / ML / GenAI:** Machine Learning, Deep Learning, CNN, LSTMs, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Agentic AI, Multi-Agent Systems, LangChain, LangGraph, LangSmith, MCP (Model Context Protocol), Transformers, Vector Databases, Agent-to-Agent (A2A) Protocol .
- **Data Analysis & BI:** Exploratory Data Analysis (EDA), Statistical Analysis, Hypothesis Testing, Data Cleaning, KPI Development, Dashboard Design, Power BI, Excel (Advanced).
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch, Transformers, FastAPI, Streamlit, SciPy, ReportLab.
- **Backend & Databases:** FastAPI, FastMCP, MongoDB, PostgreSQL, ChromaDB
- **Tools & Platforms:** GitHub, Jupyter Notebook, Vercel, Hugging Face, Linux, Bash.
- **Coursework:** Machine Learning, Probability & Statistics, Linear Algebra.

POSITION OF RESPONSIBILITY

• Active Member , AAVESH Technical Club, IIIT Una	<i>2023 – Present</i>
• Active Member , EUNOIA Literature Club, IIIT Una	<i>2023 – Present</i>